

A Certified Deterministic-Numerator Boundary Budget

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Abstract

We turn the all-order Cauchy interface of RH-252 into a usable numerical certificate. The exact RH-15 factorization and the RH-13 Arb bounds imply, without angular sampling, that the normalized Hardy-scaled deterministic numerator satisfies $M_{7/5} < 108$. The order-28 anchor of RH-253 then gives an order-29 unit-disk logarithmic tail below 0.021866475 and relative multiplicative error below 0.022107298. This certifies only the target boundary constant. It supplies neither a legal cloud head, a cloud-to-target coefficient bridge, nor a uniform quotient tail.

1 Factorization and nuclear input

Let G be the one-step deterministic numerator and let $G_H(z) = G(z/r_H)$ with $r_H = 17/20$. RH-15 proves the exact factorization [2]

$$G(u) = e^{P_1 u} \tilde{D}_{1,+}(u^2) A_*(u^2) B(u^2) C(u), \quad \tilde{D}_{1,+}(w) = \det(I - wT). \quad (1)$$

The logarithms below are the branches normalized to vanish at the origin. RH-13 realizes T on a Wiener algebra of radius $R = 0.7$ and certifies its finite section and tail with Arb arithmetic [1].

For $k \geq 1$, the nonzero reduced columns have the form

$$C_{2k} = 2a(x) \left(t(x)^k / R^{2k} - m_{2k} \right), \quad m_{2k} = \binom{2k}{k} \left(\frac{r}{2R} \right)^{2k}.$$

If $\tau = \|t\|_R / R^2 < 1$, summing the column norms gives the nuclear bound

$$\nu_1 := \nu(T) \leq 2\|a\|_R \left\{ \frac{\tau}{1-\tau} + \frac{1}{\sqrt{1-(r/R)^2}} - 1 \right\}. \quad (2)$$

Indeed, the first term sums τ^k , while the second follows from $\sum_{k \geq 0} \binom{2k}{k} x^k = (1 - 4x)^{-1/2}$.

Let M be the RH-13 finite section and let ε bound the operator remainder. Then

$$p_1 = \|M\| + \varepsilon \geq \|T\|, \quad p_2 = \|M^2\| + 2\|M\|\varepsilon + \varepsilon^2 \geq \|T^2\|. \quad (3)$$

Thus $\nu(T^2) \leq p_1 \nu_1$ and $\nu(T^3) \leq p_2 \nu_1$ by the nuclear-ideal inequality.

2 A uniform boundary theorem

Put $S = 7/5$, $U = S/r_H = 28/17$, $W = U^2$, and $t = U/\lambda$. The choice is strict: $U < \lambda$. Write $q \geq \|T^3\|$ for the RH-13 certified cube bound.

Theorem 2.1 (Modulo-three Fredholm majorant). *If $qW^3 < 1$, then throughout $|w| = W$,*

$$|\log \det(I - wT)| \leq \frac{\nu_1 W + (p_1 \nu_1) W^2 / 2 + (p_2 \nu_1) W^3 / 3}{1 - qW^3}. \quad (4)$$

Proof. For $j = 1, 2, 3$ and $k \geq 0$, the nuclear-ideal inequality gives $\nu(T^{3k+j}) \leq q^k \nu(T^j)$. Combine this with $|\operatorname{tr} T^n| \leq \nu(T^n)$ in the Fredholm logarithm, group the series modulo three, and replace $3k + j$ in the denominator by j . The remaining series is geometric with ratio qW^3 . $\square$

The endpoint factors also admit exact scalar bounds. With $b_n = \lambda^{-n}/(1 + \lambda^{-n})$ and $d_n = \lambda^{-2n}/(1 - \lambda^{-2n})$, coefficientwise absolute majorants give

$$|\log A_*(u^2)| \leq -\log(1 - t^2), \quad (5)$$

$$|\log B(u^2)| \leq \frac{-\log(1 - t^2)}{2(1 - \lambda^{-2})}. \quad (6)$$

The identity $P_1 = b_1$ cancels the linear term of $\log C$, so

$$|P_1 u + \log C(u)| \leq \operatorname{artanh}(t) - t. \quad (7)$$

Theorem 2.2 (Certified boundary budget). *For the normalized logarithm of G_H ,*

$$M_{7/5} := \sup_{|z|=7/5} |\log G_H(z)| < 107.906078 < 108.$$

Proof. The 200-decimal-place Arb replay of the RH-13 certificate gives

ν_1	< 4.623248864	p_1	< 0.633964866
p_2	< 0.174350001	qW^3	< 0.715126024
Fredholm term	< 100.715071	A_* term	< 3.291531
B term	< 2.551222	linear+ C term	< 1.348256

All entries are outward endpoints, and every displayed strict comparison is checked again at 100 and 150 decimal places. Summing (4)–(7) proves the claim. This is a uniform factor majorant on the entire circle, not a sampled maximum. $\square$

3 Order-29 consequence and boundary

RH-252 proves that Cauchy’s estimate converts M_S into an all-order target tail [3]; RH-253 supplies deterministic coefficients through order 28 [4]. Hence the first omitted order is $N = 29$.

Corollary 3.1 (Certified unit-disk target tail). *On $|z| \leq 1$,*

$$\sum_{n \geq 29} \frac{|a_n|}{n} \leq M_{7/5} \frac{(5/7)^{29}}{1 - 5/7} < 0.021866475.$$

The corresponding relative multiplicative error is below $e^{0.021866475} - 1 < 0.022107298$.

The protocol reruns the RH-13 Arb certificate at 100, 150, and 200 decimal places, evaluates only interval expressions, checks twelve outward decimal claims, and archives source hashes. It deliberately does not use the mislabelled λ^2 radius field in an older RH-46 JSON: the one-step zero-free radius used here is λ .

In the five-component RH-260 ledger [5], the status changes from $(0, 0, 0, 1, 0)$ to $(0, 0, 0, 1, 1)$. The legal anchored head, current-cloud coefficient bridge, and uniform quotient tail remain open, so the complete certificate count is zero. A finite order-28 anchor is not promoted to an all-order identification. Gates A–E remain false/open. No Hilbert–Polya operator, Riemann-zero identification, completed-zeta divisor equality, or implication of RH is asserted.

References

- [1] Bin Wang. Validated reduced-sector spectral gaps. *Preprint*, 2026. RH-13 preprint.
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